Ex: 5
Date:
30/7/26

EXPERIMENT ON PACKET CAPTURE TOOL WIRE SHARK

AIM:-

To understand the features of wireshark as a packet capture tool and understand encapsulation of information at various layers of a protocol stack.

PACKET SNIFFER:-

* Capture packets sent and received by the computer.
* Displays protocol information of captured packets.
* works passively.

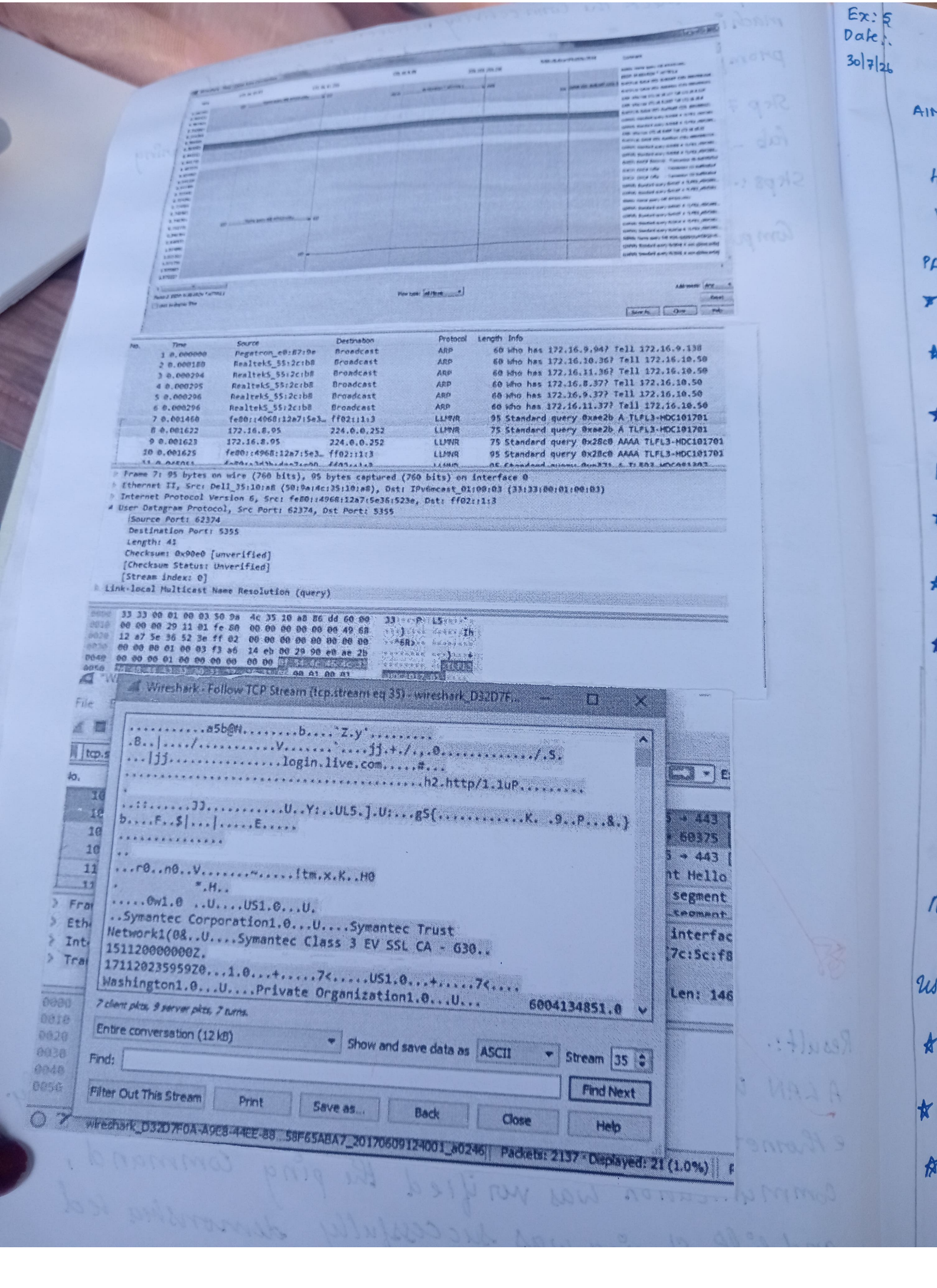
PACKET SNIFFER TOOLS:-

* TCP dump - command line packet capture tool.
* wire shark - Graphical packet analyzes.
* DESCRIPTION:-

wireshark is a network protocol analyzer that capture and displays network packets in real time. It helps monitor, analyze, troubleshoot and inspect network traffic.

Uses of Network:-

* Capture network traffic.
* Analyze protocols.
* Applies display filters.


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Wireshark · Follow TCP Stream (tcp.stream eq 35) · wireshark_D32D7F...
login.live.com
h2.http/1.1
Symantec Corporation1.0...U....Symantec Trust
Network1(0&..U....Symantec Class 3 EV SSL CA - G30..
Washington1.0...U....Private Organization1.0...U...
Entire conversation (12 kB)
Show and save data as
ASCII
Stream
35
Find:
Find Next
Filter Out This Stream
Print
Save as...
Back
Close
Help
Packets: 2137 · Displayed: 21 (1.0%)

Applications:-

* Network troubleshooting.
* Security analysis.
* Protocol debugging.
* Learning network protocols.

Capturing packet:.

1) Open wireshark.
2) Select the network interface.
3) Click start capture.
4) Observe packet in realtime.
5) Click stop and save the capture.

wireshark packet:.

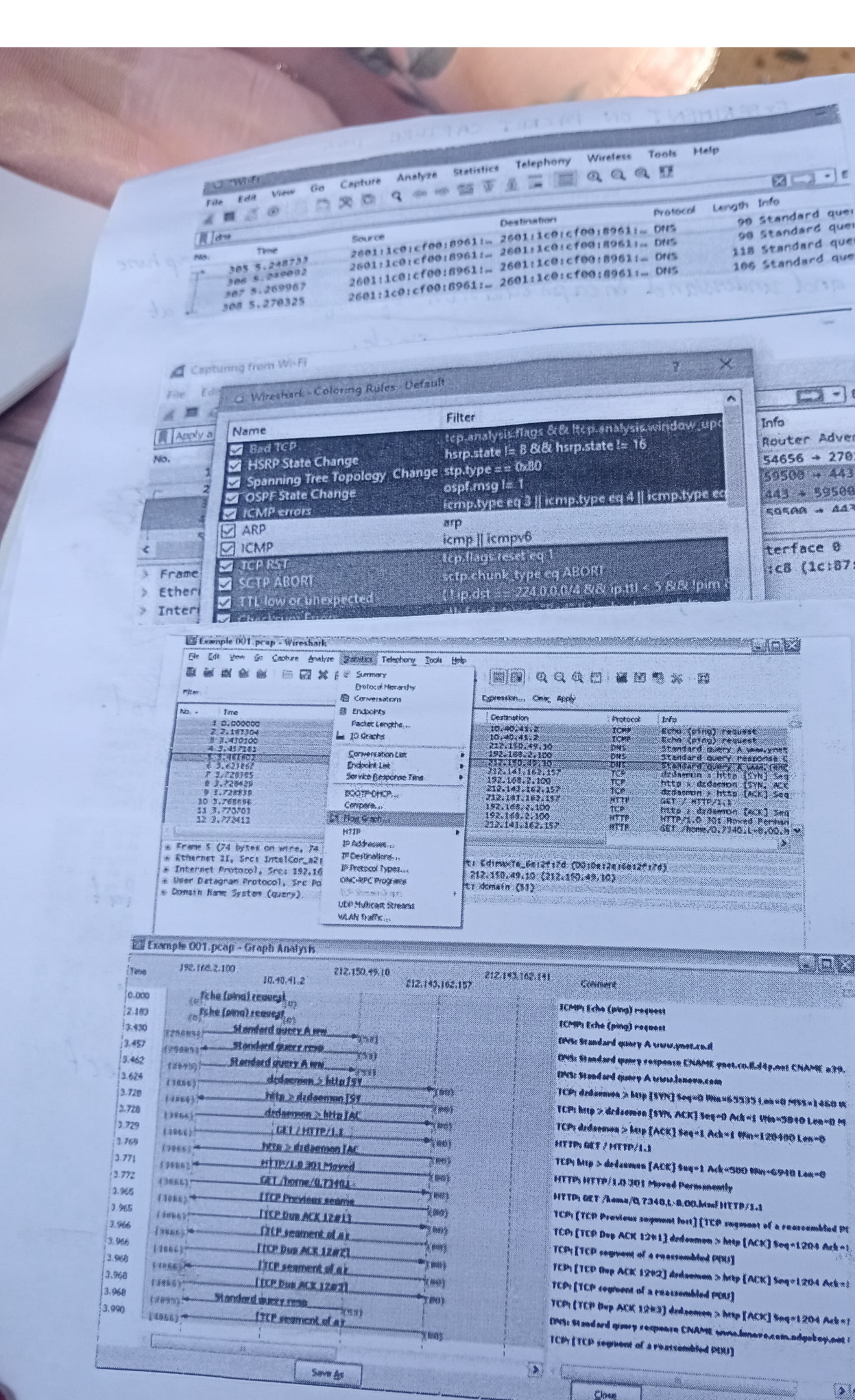
* Packet list
* Packet details.
* Packet bytes.

Color coding:-

* Purple: TCP
* Blue: VOP
* Black: Error packets.

Filtering packets:.

* Enter a filter in the filter bar.
* Right click to apply filter.


Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
No. Time Source Destination Protocol Length Info
305 5.248733 2601:1c0:cf00:8961:… 2601:1c0:cf00:8961:… DNS 90 Standard que
306 5.260092 2601:1c0:cf00:8961:… 2601:1c0:cf00:8961:… DNS 90 Standard que
307 5.269967 2601:1c0:cf00:8961:… 2601:1c0:cf00:8961:… DNS 118 Standard que
308 5.270325 2601:1c0:cf00:8961:… 2601:1c0:cf00:8961:… DNS 106 Standard que
Capturing from Wi-Fi
Wireshark · Coloring Rules · Default
Name Filter
Bad TCP tcp.analysis.flags && !tcp.analysis.window_upd
HSRP State Change hsrp.state != 8 && hsrp.state != 16
Spanning Tree Topology Change stp.type == 0x80
OSPF State Change ospf.msg != 1
ICMP errors icmp.type eq 3 || icmp.type eq 4 || icmp.type eq
ARP arp
ICMP icmp || icmpv6
TCP RST tcp.flags.reset eq 1
SCTP ABORT sctp.chunk_type eq ABORT
TTL low or unexpected (!ip.dst == 224.0.0.0/4 && ip.ttl < 5 && !pim
Info
Router Adver
54656 → 2701
59500 → 443
443 → 59500
terface 0
:c8 (1c:87:2c
Example 001.pcap - Wireshark
File Edit View Go Capture Analyze Statistics Telephony Tools Help
Summary
Protocol Hierarchy
Conversations
Endpoints
Packet Lengths...
IO Graphs
Conversation List
Endpoint List
Service Response Time
BOOTP-DHCP...
Compare...
Flow Graph...
HTTP
IP Addresses...
IP Destinations...
IP Protocol Types...
ONC-RPC Programs
UDP Multicast Streams
WLAN Traffic...
Destination Protocol Info
10.40.41.2 ICMP Echo (ping) request
10.40.41.2 ICMP Echo (ping) request
212.150.49.10 DNS Standard query A www.ynet.co.il
192.168.2.100 DNS Standard query response C
212.150.49.10 DNS Standard query A www.lenovo.com
212.141.162.157 TCP drdaemon > http [SYN] Seq
192.168.2.100 TCP http > drdaemon [SYN, ACK
212.141.162.157 TCP drdaemon > http [ACK] Seq
212.141.162.157 HTTP GET / HTTP/1.1
192.168.2.100 TCP http > drdaemon [ACK] Seq
192.168.2.100 HTTP HTTP/1.0 301 Moved Permanen
212.141.162.157 HTTP GET /home/0,7340,L-8,00.h
Example 001.pcap - Graph Analysis
Time 192.168.2.100 10.40.41.2 212.150.49.10 212.141.162.157 212.143.162.141 Comment
0.000 Echo (ping) request
2.183 Echo (ping) request
3.430 Standard query A ww
3.457 Standard query resp
3.462 Standard query A ww
3.624 drdaemon > http [SY
3.728 http > drdaemon [SY
3.728 drdaemon > http [AC
3.729 GET / HTTP/1.1
3.769 http > drdaemon [AC
3.771 HTTP/1.0 301 Moved
3.772 GET /home/0,7340,L-
3.965 [TCP Previous segme
3.965 [TCP Dup ACK 1201#1]
3.966 [TCP segment of a r
3.966 [TCP Dup ACK 1202#2]
3.968 [TCP segment of a r
3.968 [TCP Dup ACK 1203#3]
3.990 Standard query resp
[TCP segment of a r
ICMP: Echo (ping) request
ICMP: Echo (ping) request
DNS: Standard query A www.ynet.co.il
DNS: Standard query response CNAME ynet.co.il.dip.net CNAME a39.
DNS: Standard query A www.lenovo.com
TCP: drdaemon > http [SYN] Seq=0 Win=65535 Len=0 MSS=1460 W
TCP: http > drdaemon [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 M
TCP: drdaemon > http [ACK] Seq=1 Ack=1 Win=128480 Len=0
HTTP: GET / HTTP/1.1
TCP: http > drdaemon [ACK] Seq=1 Ack=580 Win=6948 Len=0
HTTP: HTTP/1.0 301 Moved Permanently
HTTP: GET /home/0,7340,L-8,00.html HTTP/1.1
TCP: [TCP Previous segment lost] [TCP segment of a reassembled PD
TCP: [TCP Dup ACK 1201#1] drdaemon > http [ACK] Seq=1204 Ack=1
TCP: [TCP segment of a reassembled PDU]
TCP: [TCP Dup ACK 1202] drdaemon > http [ACK] Seq=1204 Ack=1
TCP: [TCP segment of a reassembled PDU]
TCP: [TCP Dup ACK 1203] drdaemon > http [ACK] Seq=1204 Ack=1
DNS: Standard query response CNAME www.lenovo.com.edgekey.net
TCP: [TCP segment of a reassembled PDU]
Save As
Close

FLOW GRAPH:-

Displays communication flow between devices, making packing exchange easier to understand

Capture 100 packets.

ii) Open wireshark

Create a filter to display only DNS packets and provide the flow graph.

-> Go to Capture option.

-> select stop capture automatically after 100 packets.

-> Then click start capture.

-> Search DNS packets in search bar.

-> To see flow graph click statistics flow graph.

-> Save the packets.

Create a filter to display only HTTP packets and inspect the packets procedure.

-> Select Local Area Connection in wireshark.

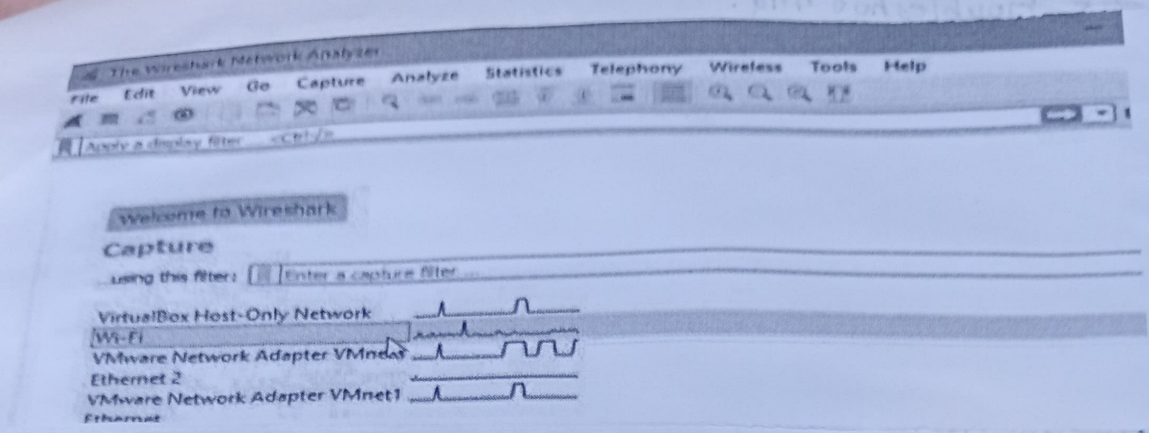
-> Go to Capture option.

-> Select stop capture automatically after 100 results.

-> Then click start capture.

-> search ICMP/IP packets in search bar.

-> Save the packets.



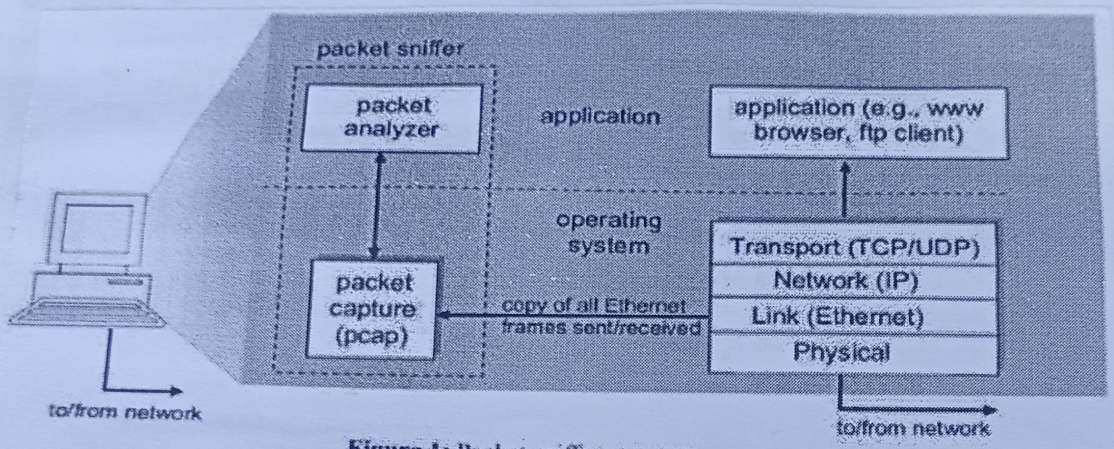
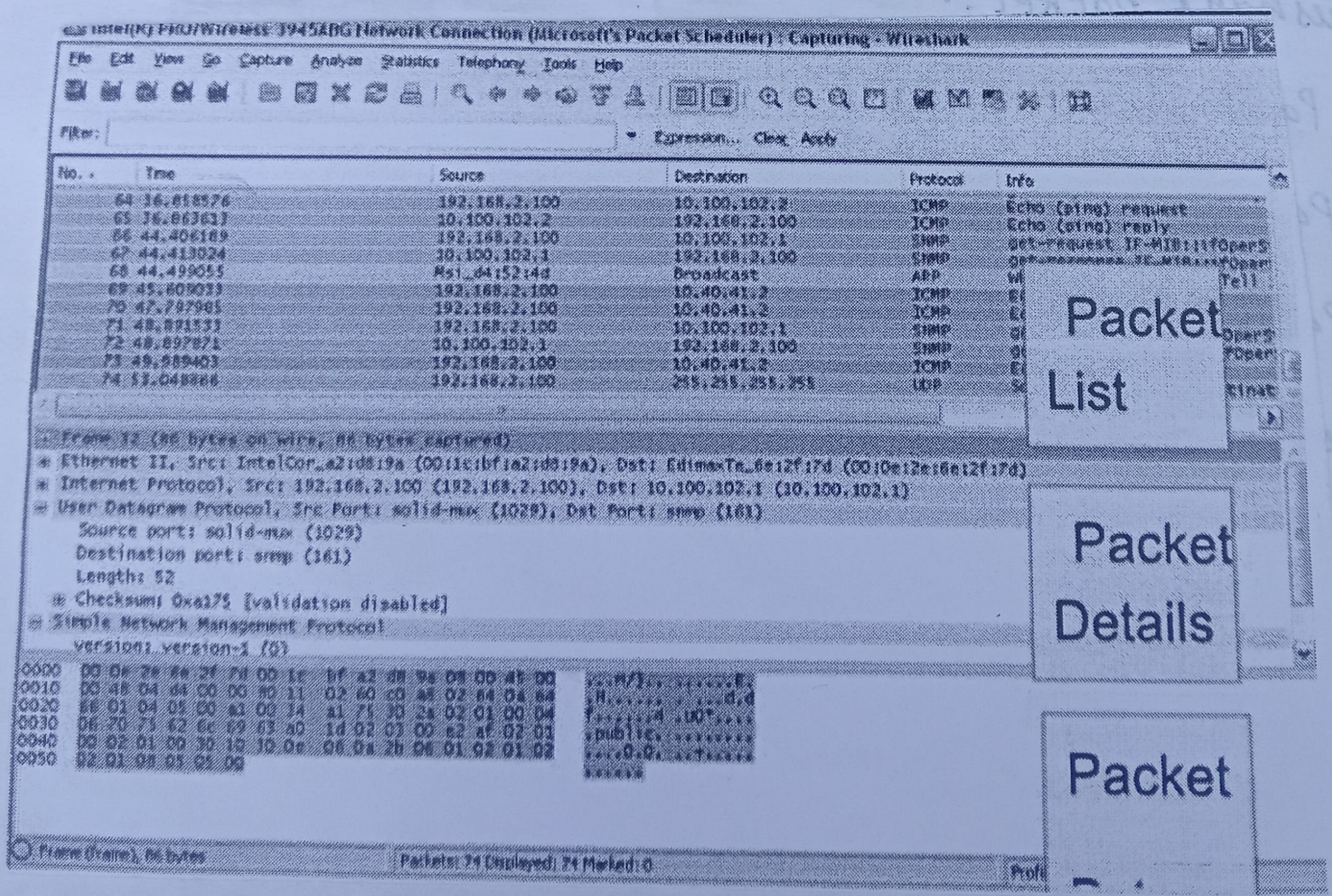


Figure 1: Packet sniffer structure



Create a filter to display only DHCP packets and inspect the packets.

Procedure:

-> Select local area connection in wireshark.
-> Go to capture option.
-> Select stop capture automatically after 100 packets.
-> Then click start capture.
-> Search DHCP packets in search bar.
-> Save the packets.

RESULT:-

Thus the experiments are excuted successfully and experiments on packet capture tool wiresshark.